

Universal Sales & Pricing Analytics Tool

How it works — explained in plain language

This document explains, without technical jargon, what the tool does, how it is built, what it looks like, and how its correctness was verified.

Live app: inigoserrano-sales-pricing-analytics.streamlit.app · Source code: github.com/inigosueras/sales-pricing-analytics

1. What is this tool?

It is a program that turns a sales file (Excel or CSV) into a complete commercial analytics dashboard, automatically. No configuration is needed: you upload the file and the tool calculates the key metrics, trends, best-selling products, breakdown by country and by customer segment, and more, on its own.

The key feature is that it works equally well with data from completely different industries (retail, pharma, manufacturing...) because it does not depend on specific product or category names — it only requires the file to follow a fixed column structure.

Columns the input file must have:

Column	What it represents
Date	Date of the sale
Product	Name of the product sold
Category	Category the product belongs to
Country	Country where the sale took place
Revenue	Revenue generated by that sale
Quantity	Units sold
Cost	Cost associated with that sale

Customer_Segment	Customer segment (e.g. Online, Wholesale, Hospital...)
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2. How does it work under the hood?

The tool is organized into 3 layers, like the floors of a building. Each one does a specific job and builds on the one below it:

Layer 1 — The data engine

The invisible part that does the heavy lifting, in 5 automatic steps:

- 1. Validation:** checks that the file has the correct columns and that the data makes sense (for example, no negative revenue or zero quantities), pointing to exactly which column and row has a problem when something is wrong.
- 2. Category cleanup:** near-duplicate text values (e.g. "Online", "on-line", "ONLINE ") are automatically merged into a single label, using exact-match normalization plus optional fuzzy matching for typos — without merging genuinely distinct short codes like "US" vs. "UK".
- 3. Type conversion and calculation:** from the basic columns, it computes margin, margin %, average price, unit cost, and the year/quarter/month of each sale, plus the year-over-year (YoY) comparison.
- 4. Storage:** saves the cleaned data into a lightweight database (SQLite), allowing fast queries even with hundreds of thousands of rows.
- 5. Multiple files at once:** every file you upload is tagged with its own identity, so you can have several loaded at the same time (for example, retail and pharma) and compare them against each other, without them mixing or overwriting one another.

Layer 2 — The web application

The visual part: a web page where you upload your file and the key metrics, charts and tables appear automatically, with filters to narrow down by period, country, category, segment, or dataset. Filters behave like a Power BI slicer: selecting or deselecting a dataset only changes what is shown, it never deletes anything, and the selection is remembered as you move between pages.

Layer 3 — Power BI template

A version of the same analysis inside Power BI: the M script (data loading and cleaning) and DAX measures (KPIs, YoY, price/volume) are complete and included in the repository, along with a step-by-step build guide, ready to be assembled into a .pbit template in Power BI Desktop.

3. What exactly do you get from the tool?

Once a valid file is uploaded, the tool automatically generates:

- **Main KPIs:** total revenue, margin %, average ticket, units sold, distinct products and countries.
- **Monthly / quarterly trend** with year-over-year (YoY) comparison.
- **Top 10 products and categories** by revenue, margin, or units.
- **Geographic distribution:** a map (automatically zoomed to the region with data) and a table by country.
- **Price vs. volume decomposition:** how much of a revenue change is due to selling at a higher price, and how much to selling a higher quantity.
- **Customer segment analysis:** revenue share and margin by segment.
- **Pivot Explorer:** freely combine 1 to 3 dimensions (e.g. Country + Category + Product), visualized as a treemap — the equivalent of dragging fields into an Excel PivotTable.
- **Multi-dataset comparison:** load several files at once and compare their KPIs side by side.

4. Walkthrough

The following screenshots are taken directly from the live, deployed application.

The screenshot shows the home page of the 'Universal Sales & Pricing Analytics Tool'. On the left is a sidebar with navigation links: Overview, Trends, Products Categories, Geography, Price Volume, Segments, and Explorer. The main content area has a header with the tool's name and a brief instruction. Below this is a 'Quick start with sample data' section with three buttons: 'Load Retail sample', 'Load Pharma sample', and 'Load Manufacturing sample'. Further down is an 'Upload your own file' section with a text input for a dataset label and an 'Upload' button. At the bottom, a 'Datasets loaded in this session' section shows a table with one entry: 'retail_sales.csv' (Retail, 6,000 rows, loaded on 2026-08-20 13:19:51). A 'Manage app' button is in the bottom right corner.

streamlit app

Overview
Trends
Products Categories
Geography
Price Volume
Segments
Explorer

Universal Sales & Pricing Analytics Tool

Upload a sales file with the required column structure and get a complete commercial analytics dashboard — automatically, no configuration needed.

Required column structure

Quick start with sample data

Load Retail sample Load Pharma sample Load Manufacturing sample

Upload your own file

Label for this dataset (optional)
e.g. Retail EU, Q3 Pharma...

CSV or Excel file

Upload 250MB per file • CSV, XLSX, XLS

Datasets loaded in this session

Dataset Name	Category	Rows	Loaded At
retail_sales.csv	Retail	6,000 rows	2026-08-20 13:19:51

Use the pages in the sidebar to explore Overview, Trends, Products & Categories, Geography, Price/Volume, and Segments. Use the filter panel on each page to select which dataset(s) to analyze.

Manage app

Home — upload a file or load one of the three sample datasets (retail, pharma, manufacturing) in one click.

The screenshot shows the 'Overview' page. The sidebar is updated with 'Overview' selected. The main content area features a 'Revenue Trend Snapshot' section. At the top, four key metrics are displayed: Total Revenue (\$13,856,810), Margin % (50.6%), Avg Ticket (\$2,309.47), and Units Sold (140,146). Below these, two more metrics are shown: Total Margin (\$7,011,338) and Distinct Products (8). A line chart titled 'Revenue Trend — Current vs. Prior Year' shows revenue from Jan 2023 to Oct 2024. The chart compares current revenue (solid blue line) with prior year revenue (dashed blue line). The y-axis ranges from \$450,000 to \$650,000. The x-axis shows quarterly intervals. A 'Filters' sidebar on the left allows filtering by Dataset (Retail), Date range (2023/01/01 to 2024/12/31), Country, Category, and Customer Segment. A 'Manage app' button is in the bottom right corner.

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Overview
Trends
Products Categories
Geography
Price Volume
Segments
Explorer

Overview

Total Revenue
\$13,856,810

Margin %
50.6%

Avg Ticket
\$2,309.47

Units Sold
140,146

Total Margin
\$7,011,338

Distinct Products
8

Distinct Countries
6

Revenue Trend Snapshot

Revenue Trend — Current vs. Prior Year

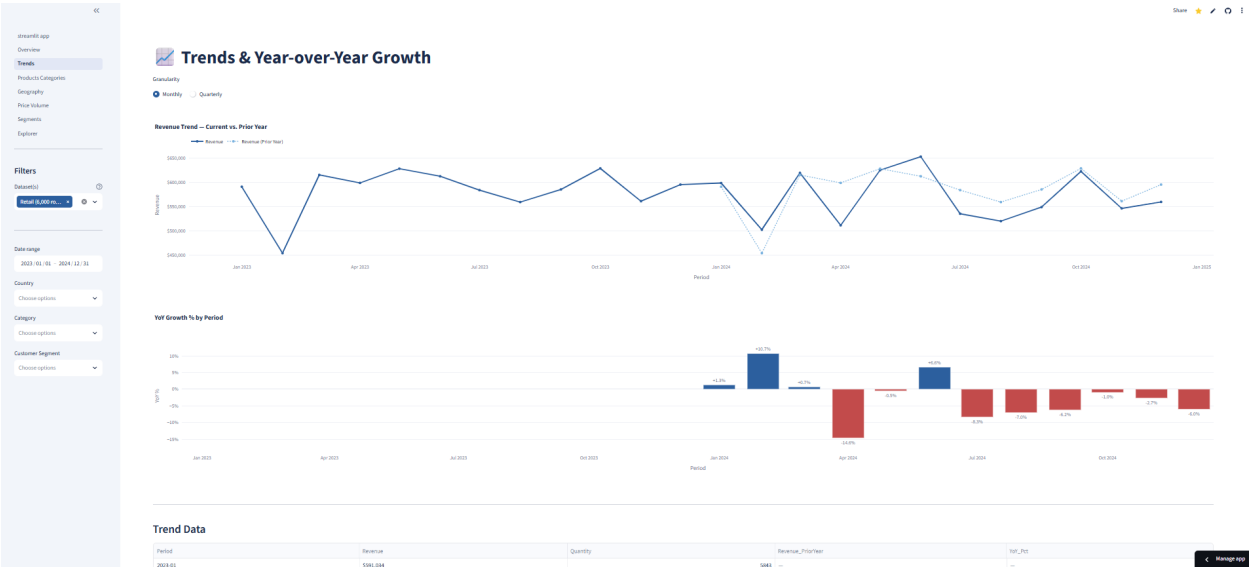
Revenue

Period

For quarterly view, YoY growth %, and full trend breakdown, see the Trends page.

Manage app

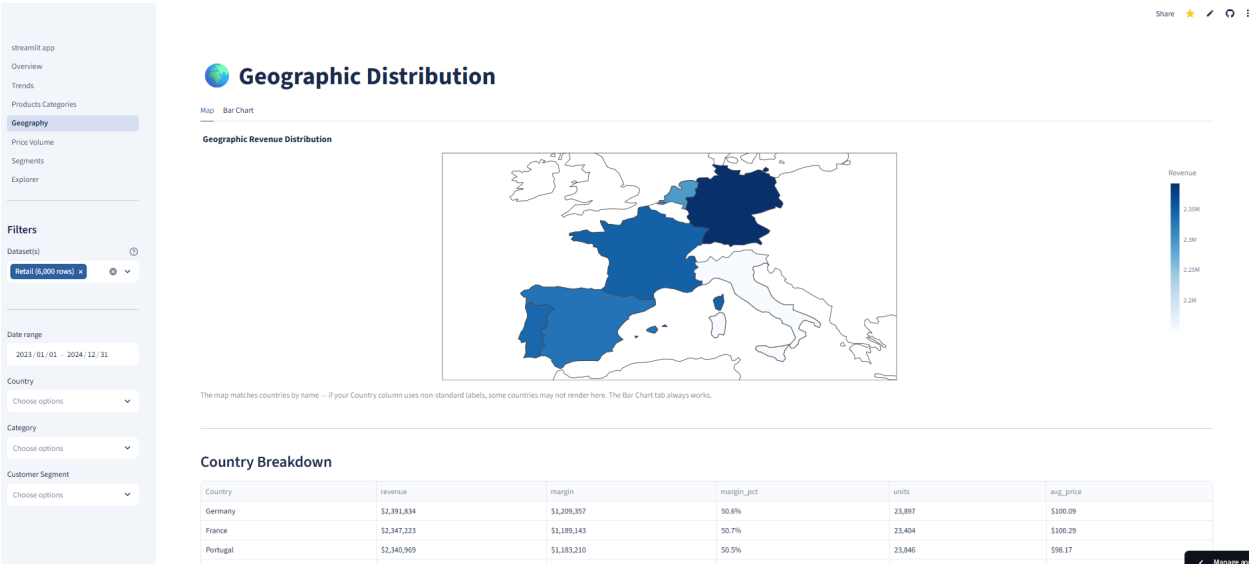
Overview — main KPIs and a revenue trend snapshot, filtered by the datasets and periods selected in the sidebar.



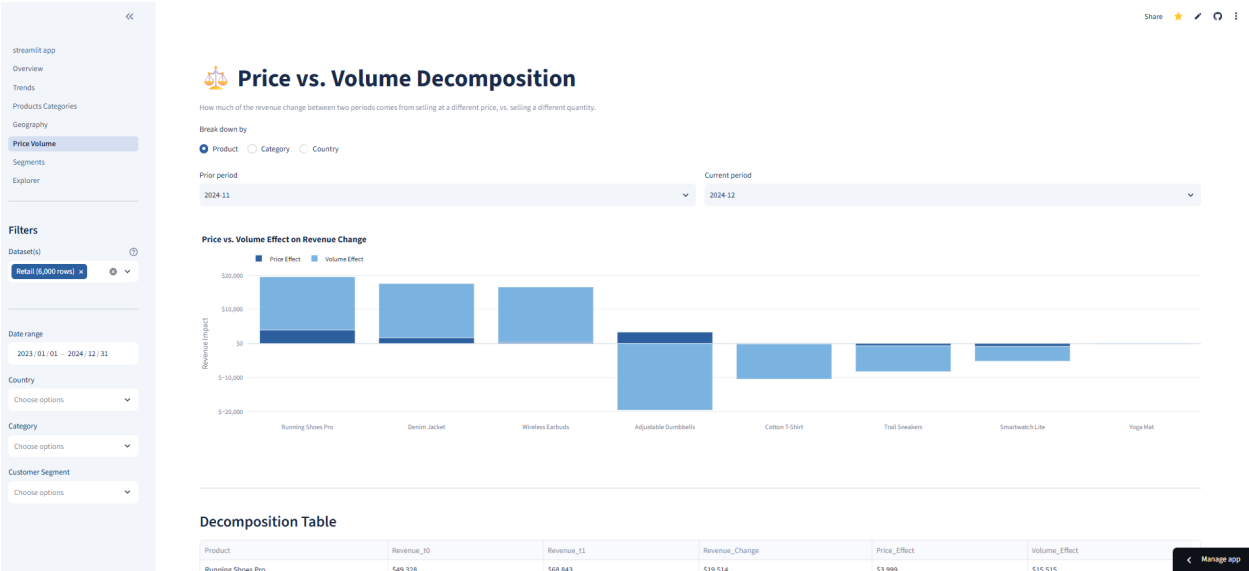
Trends — monthly or quarterly revenue vs. prior year, with a YoY growth % chart color-coded by direction.



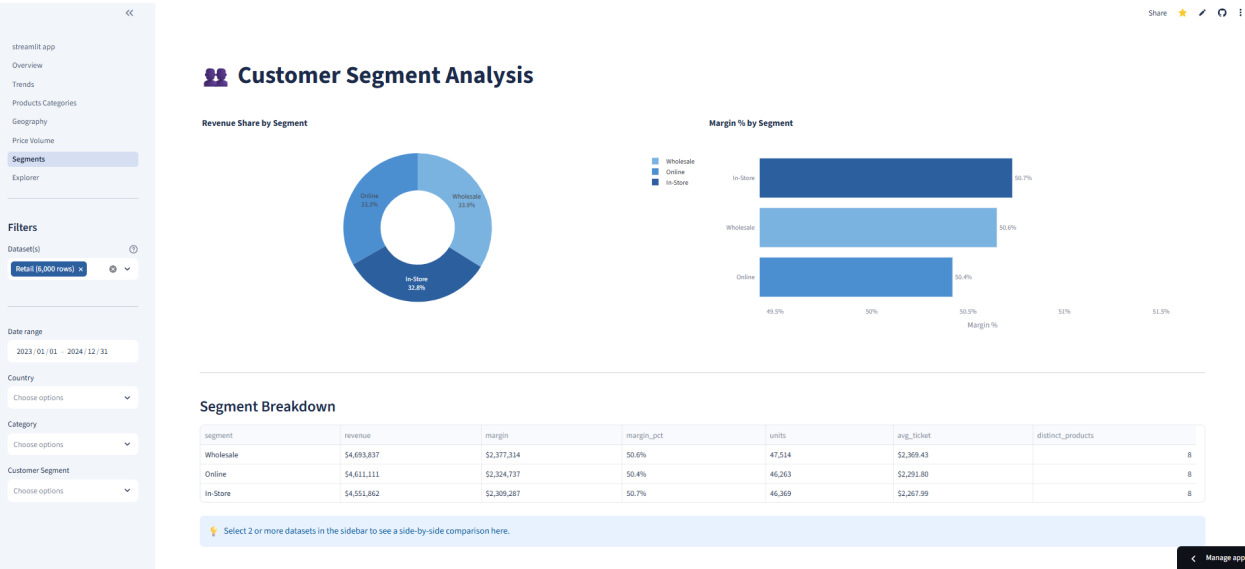
Products & Categories — Top 10 ranking with value labels, sortable by revenue, margin, or units.



Geography — the map auto-zooms to the region containing data, instead of showing a mostly-empty world map.



Price vs. Volume — decomposes the revenue change between two periods into a price effect and a volume effect.



5. How is correct behavior verified?

The reliability of the tool is verified at three levels, from most technical to most visual:

Level 1 — Automated tests

Each part of the code is given a known input and the output is checked against the exact expected result. There are **32 tests, and all 32 pass**. They cover file validation, category normalization (including a safety check that short codes like "US"/"UK" are never falsely merged), financial calculations, database loading, filters, and protection against malicious data input — including the dynamic multi-dimension queries behind the Pivot Explorer.

Level 2 — Testing with real data from 3 industries

Three sample files were generated (retail, pharma, manufacturing) and run through the full pipeline end to end. All three worked without changing a single line of code, demonstrating that the tool is genuinely universal rather than built around one specific case.

Dataset	Rows	Total revenue	Margin %
Retail	6,000	\$13,856,809.82	50.6%
Pharma	7,000	\$3,281,129.07	57.8%
Manufacturing	5,000	\$48,555,035.82	44.9%

Level 3 — Live deployment

The app is deployed and publicly accessible at the link on the cover page. Anyone can load a sample dataset or upload their own file and see the metrics and charts build in real time — this document's screenshots (Section 4) were taken directly from that live deployment.

6. Summary

Deliverable	Status
Python data engine (validation, cleaning, calculation, database)	Completed and tested
3 sample datasets (retail, pharma, manufacturing)	Completed
Interactive web application (Streamlit, 7 pages)	Completed and deployed
Power BI template materials (M script, DAX, build guide)	Completed
GitHub repository with documentation	Completed

Online deployment (Streamlit Cloud)	Completed
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Source code: github.com/inigosueras/sales-pricing-analytics

Built by **Iñigo Serrano** — BI & Pricing Analytics professional.